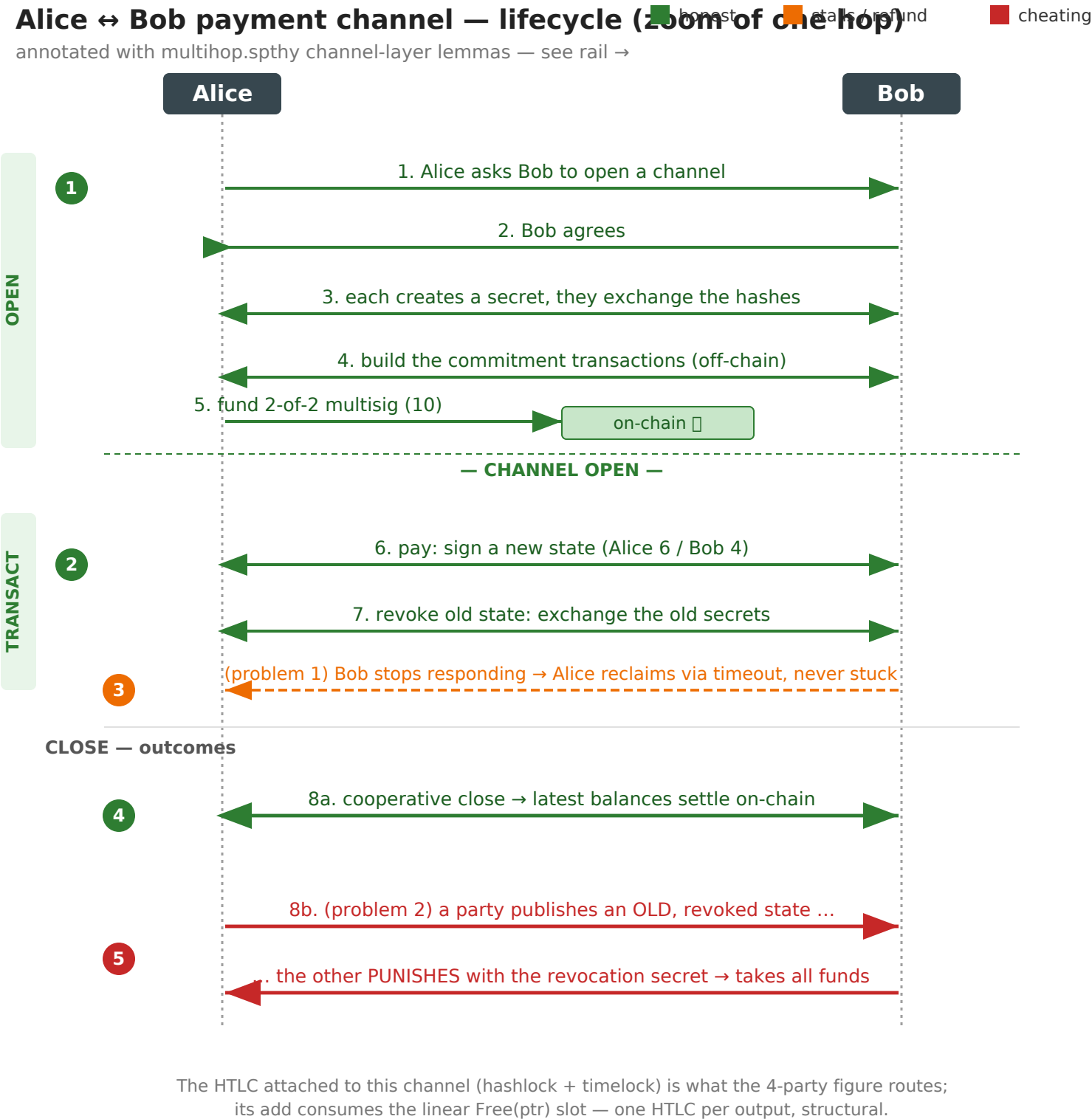


Alice ↔ Bob payment channel — lifecycle (zoom of one hop)

annotated with multihop.spthy channel-layer lemmas — see rail →



LEMMAS · channel layer

- 1 **Channel open & funding**
Protocol_execution ✓ (witness) · Distinct_Parties_Configuration ✓
Funds_Locked_Before_Update ✓ · HTLC_On_Opened_Channel ✓
- 2 **State update & revocation**
state_update ✓ · Update_Requires_Negotiation ✓
Invoice_Released_Once ✓ · Ltk_Known_Implies_Compromised ✓
- 3 **Refund / anti-stall**
Refund_Requires_Timeout ✓ · Refund_Possible ✓ (witness)
- 4 **Cooperative close & settlement**
Cooperative_Close_Execution ✓ · settlement_is_traceable ✓
instant_funds ✓ · delayed_funds ✓
- 5 **Punishment (safety, not an attack)**
No_Punishment_Without_Cheating ✓
— red trace = the cheat; the lemma proves punishment ⇒ prior

✓ = machine-checked in multihop.spthy.
Routing / value / attack lemmas are on the 4-party figure (anchors ①–⑧ there).